1. 同步 Buck-Boost 升降压输出
2. DAC控制输出电压
3. 输出电压闭环控制
4. 输出电流检测
5. 限流 / 恒流控制
6. 短路保护
7. 输入欠压保护
8. 输出过压保护（未设计）
9. 过温保护
10. 软启动
11. 输出使能控制
12. 输出电压快速跟踪
13. 工作状态监测
14. 故障状态指示
15. 预留 MCU / 通信 / MQTT 扩展接口

全工作输入电压：16V ～ 20V，标称值 18V

输出电压：0V ～ 35V

30V 输出时，输出电压纹波 < 250mV

跟踪功能：30V 降至 5V 的响应时间为 2000μs

同步频率控制范围：400kHz ～ 500kHz

设计目标：系统成本低于现有方案

峰值输出功率：300W（第一版）

波峰因数（有效值与峰值之比）：4（对应 75W 持续功率）

峰值功率维持时间：200ms

标称效率：满载、标称输入电压下 > 93%

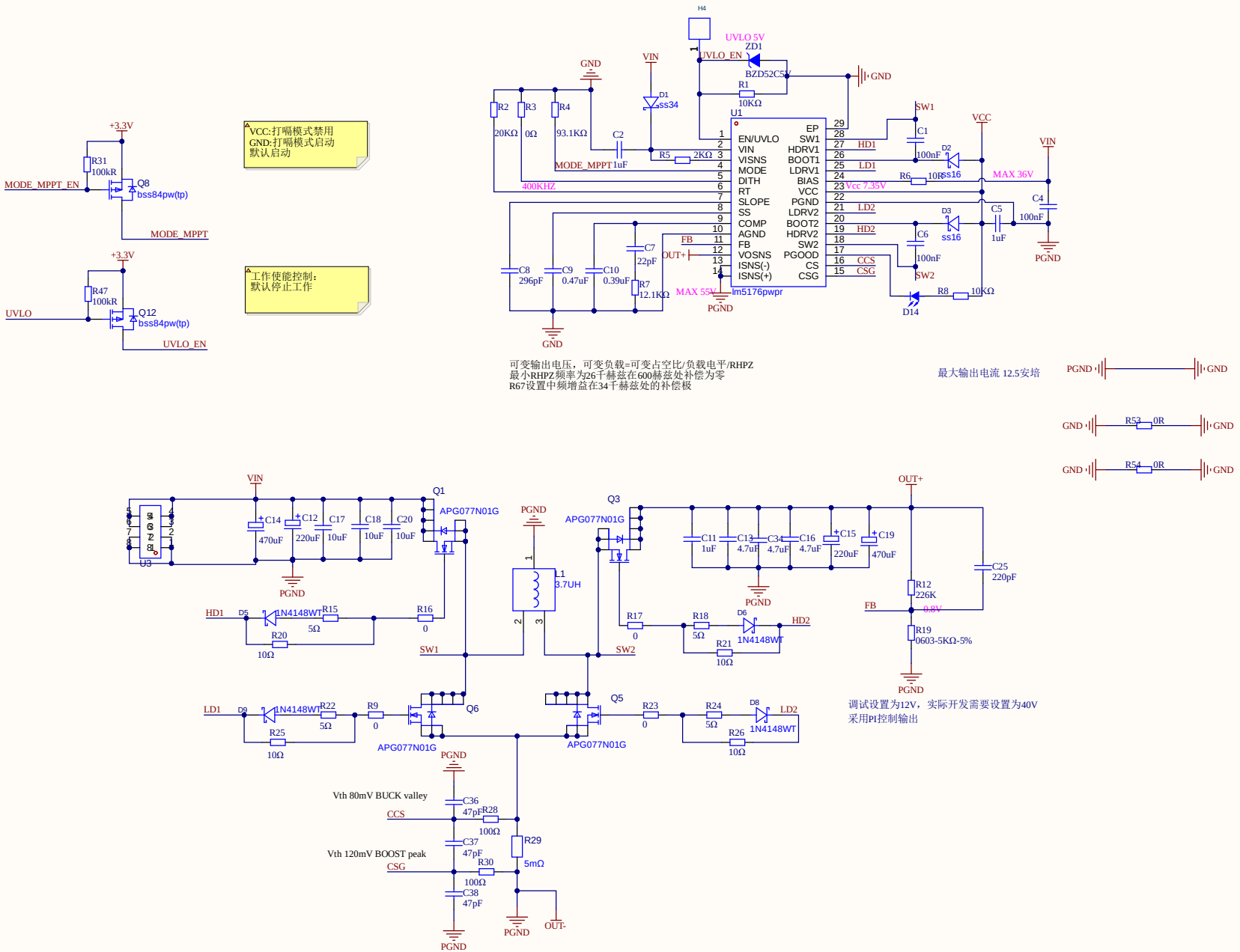
采用 DAC控制输出电压

最高环境工作温度：85°C

具备短路保护

具备限流功能

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Date: 10/28/2026		Sheet of
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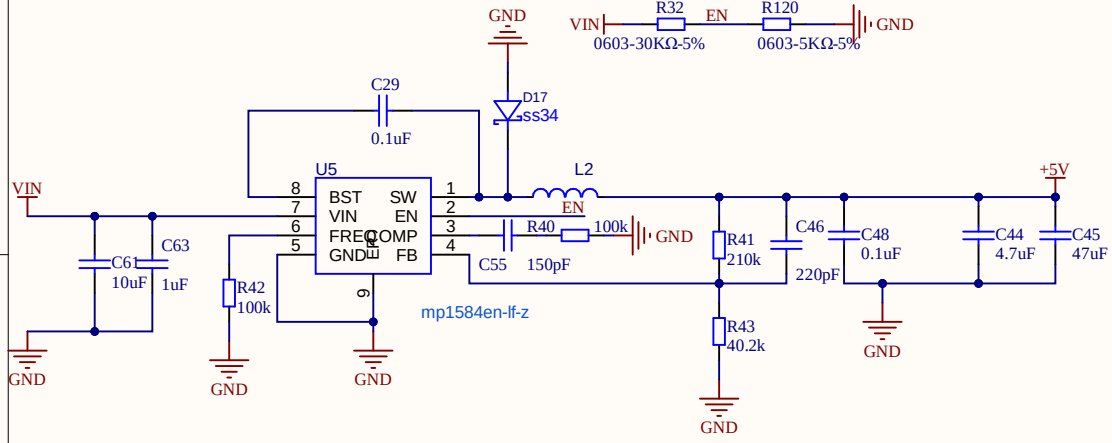


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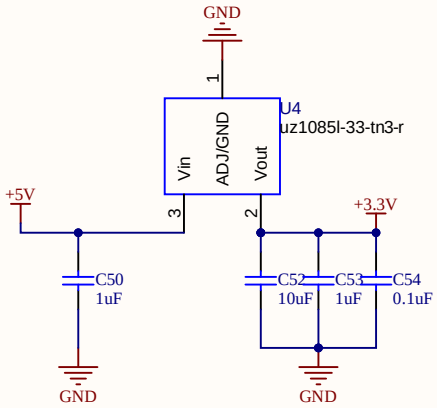
DC-DC模块/转+5V

默认突发模式
VCC:脉冲跳跃模式
GND:Burst Mode®
(突发模式)

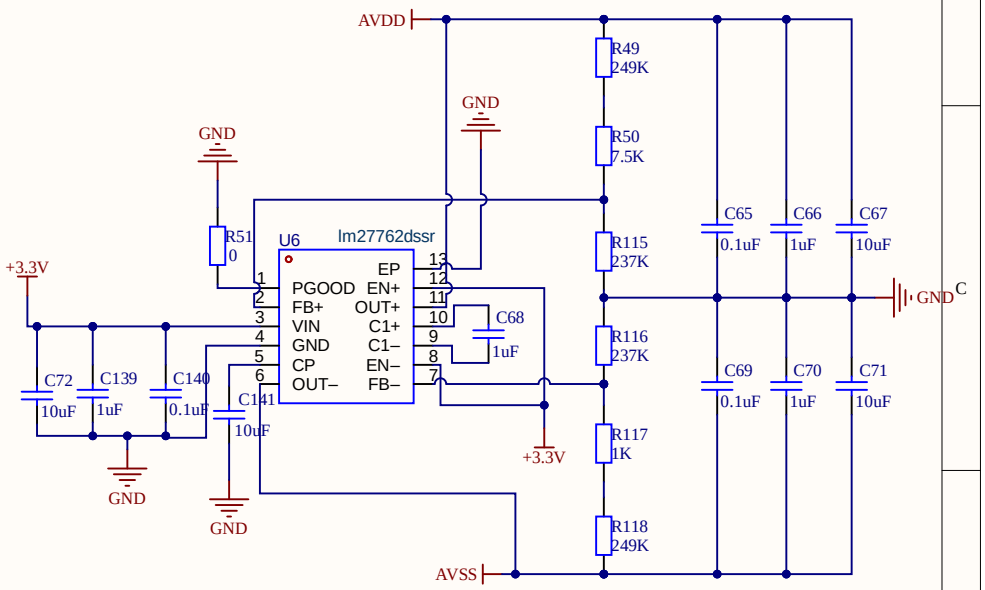
输入范围支持 3.4V ~ 42V
输出电压: 5V
最大输出电流：4A
开关频率 RRT = 110kΩ，对应约 400kHz
反馈基准 约 0.97V
反馈分压 1MΩ / 243kΩ
计算输出 0.97 x (1 + 1M / 243k) ≈ 4.96V



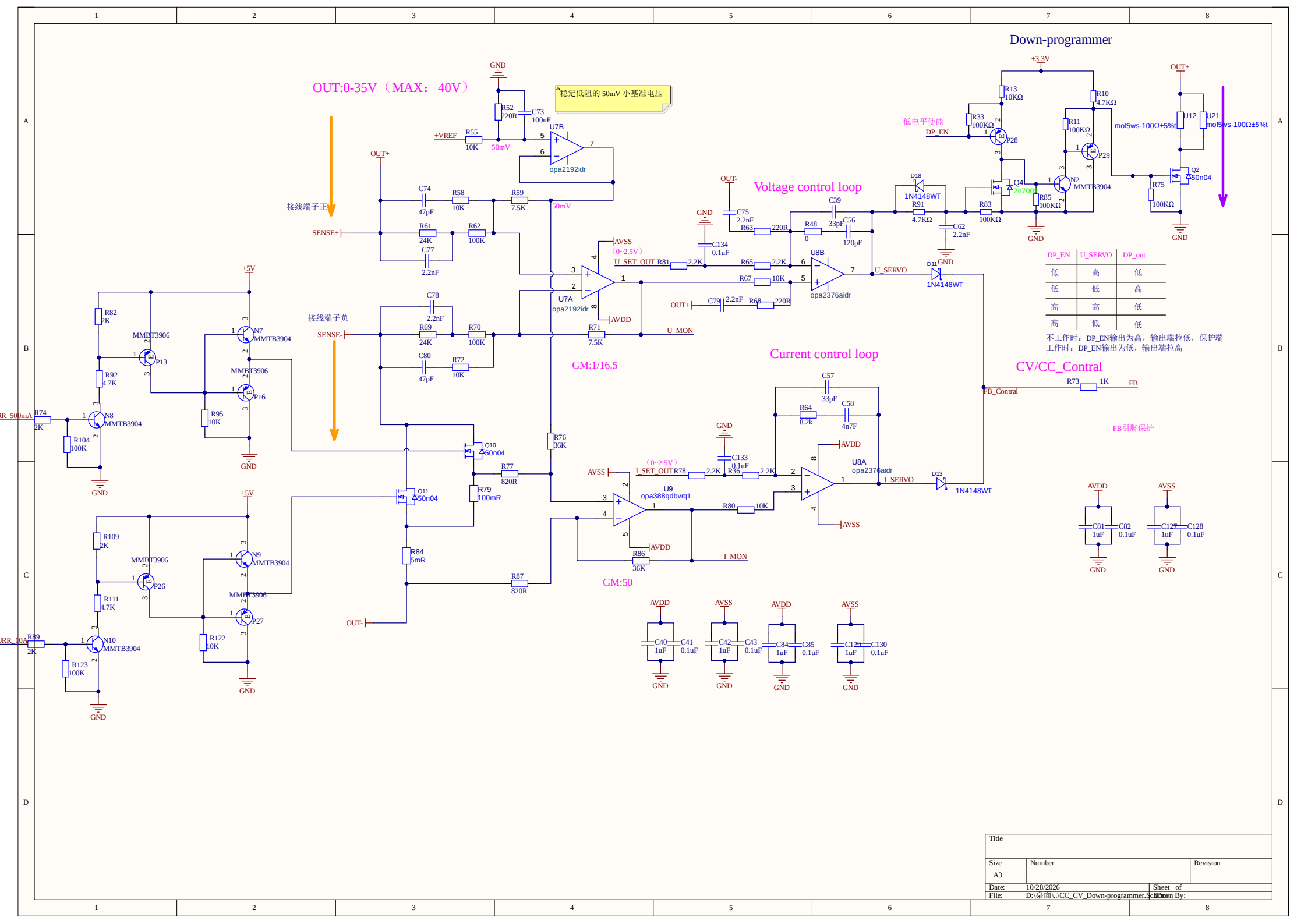
LDO模块/转+3.3V



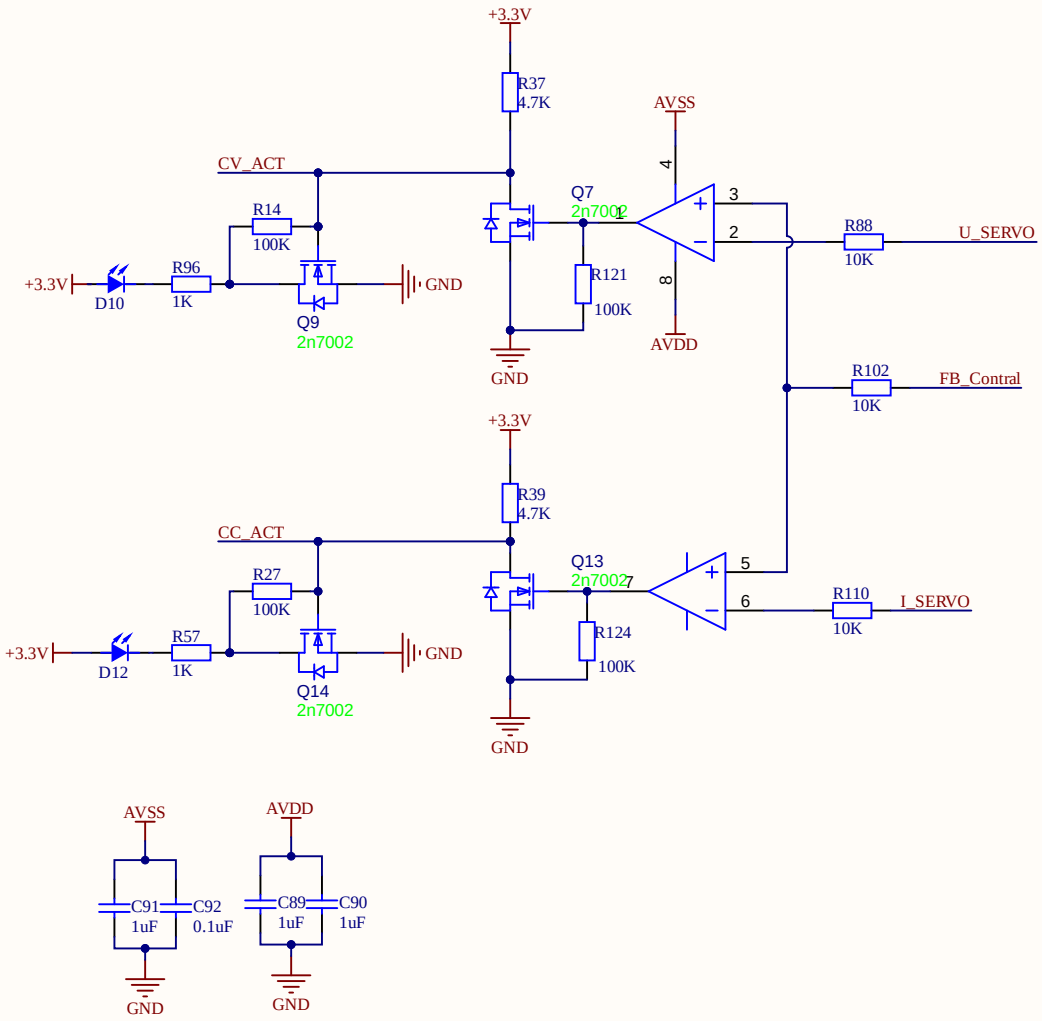
DC-DC模块/+5转+-2V5



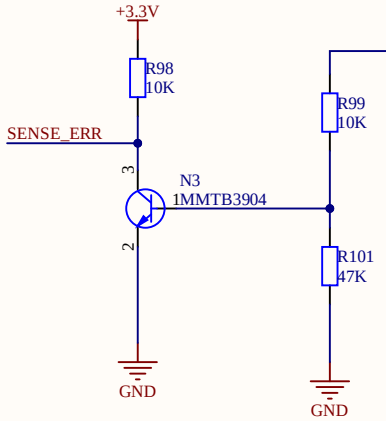
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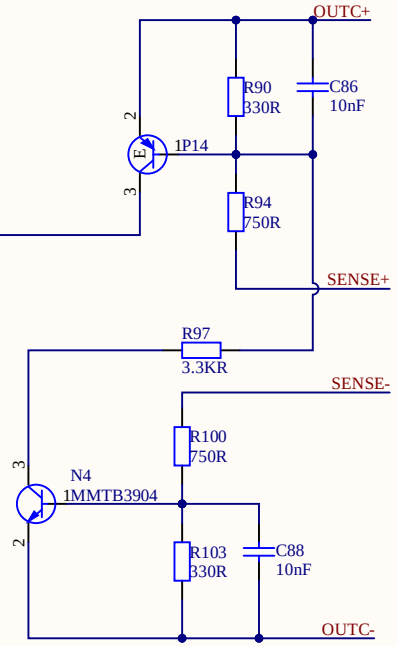

Down-programmer
OUT:0-35V（MAX：40V）
稳定低阻的 50mV 小基准电压
接线端子正
接线端子负
SENSE+
SENSE-
OUT+
OUT-
+VREF
R55 10K
50mV
R52 220R
C73 100nF
U7B
opa2192idr
C74 47pF
R58 10K
R59 7.5K
R61 24K
R62 100K
C77 2.2nF
C78 2.2nF
R69 24K
R70 100K
C80 47pF
R72 10K
U7A
opa2192idr
AVSS
（0~2.5V）
U_SET_OUT R81
2.2K
AVDD
R71 7.5K
U_MON
GM:1/16.5
Voltage control loop
C75 2.2nF
R63 220R
C39 33pF
C56 120pF
R48 0
C134 0.1uF
R65 2.2K
R67 10K
U8B
opa2376aidr
U_SERVO
C79 2.2nF
R68 220R
D18 1N4148WT
R91 4.7KΩ
C62 2.2nF
D11 1N4148WT
低电平使能
DP_EN
R13 10KΩ
R33 100KΩ
P28
R10 4.7KΩ
R11 100KΩ
P29
Q4 2n7002
N2 MMTB3904
R85 100KΩ
R83 100KΩ
mof5ws-100Ω±5%t
U12
U21
Q2 50n04
R75 100KΩ
DP_EN | U_SERVO | DP_out
低 | 高 | 低
低 | 低 | 高
高 | 高 | 低
高 | 低 | 低
不工作时：DP_EN输出为高，输出端拉低，保护端
工作时：DP_EN输出为低，输出端拉高
Current control loop
C57 33pF
R64 8.2k
C58 4n7F
C133 0.1uF
I_SET_OUTR78
R36 2.2K
U8A
opa2376aidr
I_SERVO
D13 1N4148WT
U9
opa388qdbvrq1
R80 10K
R86 36K
I_MON
GM:50
CV/CC_Contral
R73 1K
FB
FB_Contral
FB引脚保护
R76 36K
R77 820R
R79 100mR
R84 5mR
R87 820R
Q10 50n04
Q11 50n04
R82 2K
MMBT3906
N7 MMTB3904
P13
R92 4.7K
P16
R95 10K
R74 2K
N8 MMTB3904
R104 100K
R109 2K
P26
N9 MMTB3904
R111 4.7K
P27
R122 10K
R89 2K
N10 MMTB3904
R123 100K
+5V
GND
C40 1uF
C41 0.1uF
C42 1uF
C43 0.1uF
C84 1uF
C85 0.1uF
C129 1uF
C130 0.1uF
C81 1uF
C82 0.1uF
C127 1uF
C128 0.1uF
+3.3V
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恒压（CV）和恒流（CC）模式指示



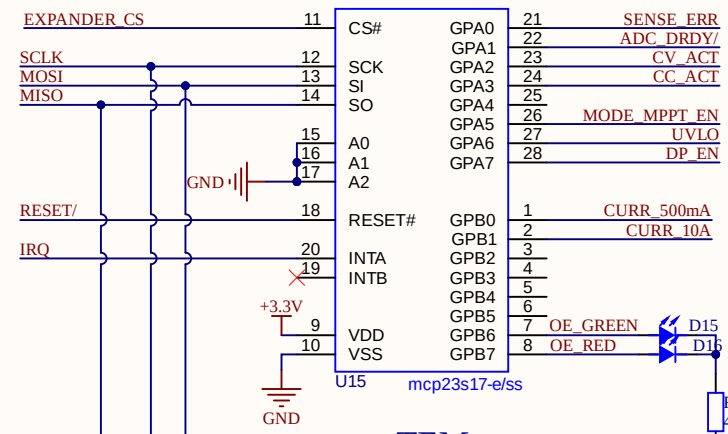
远端极性反接检测



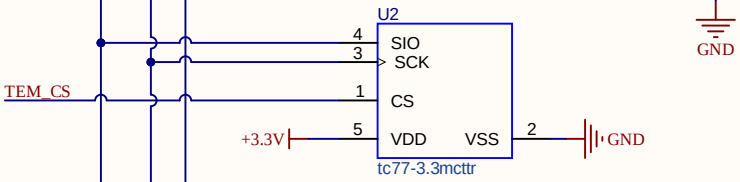


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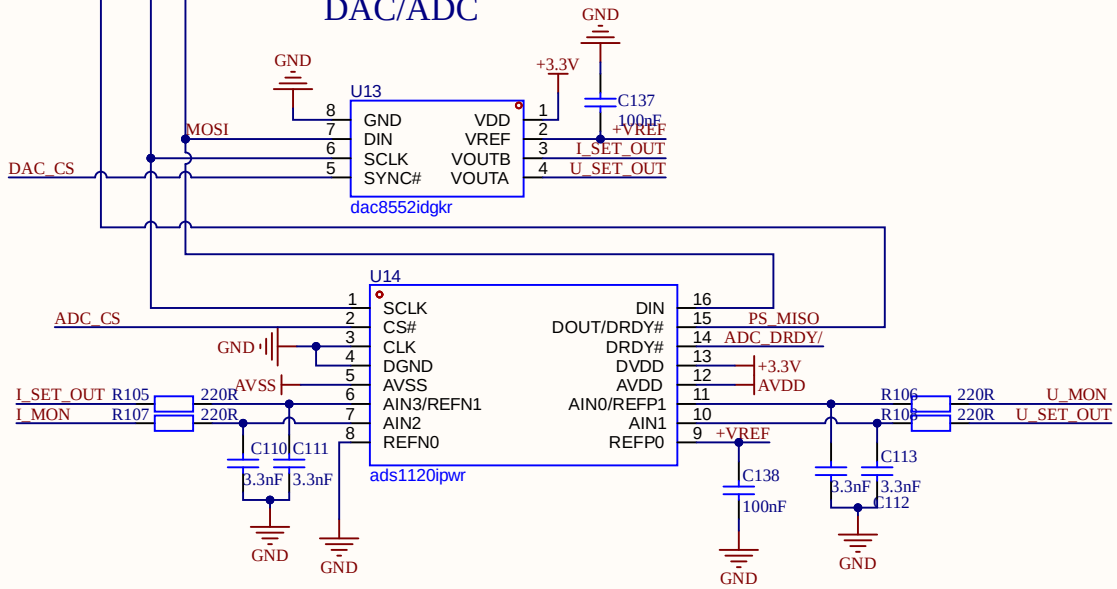
SPI_IO_EXPANDER



TEM



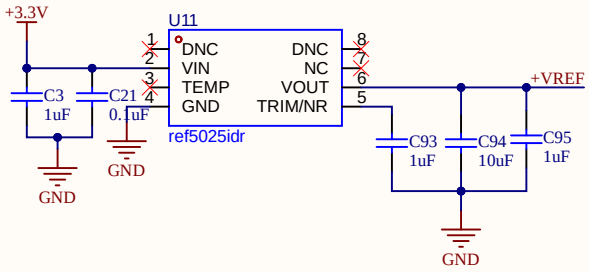
DAC/ADC

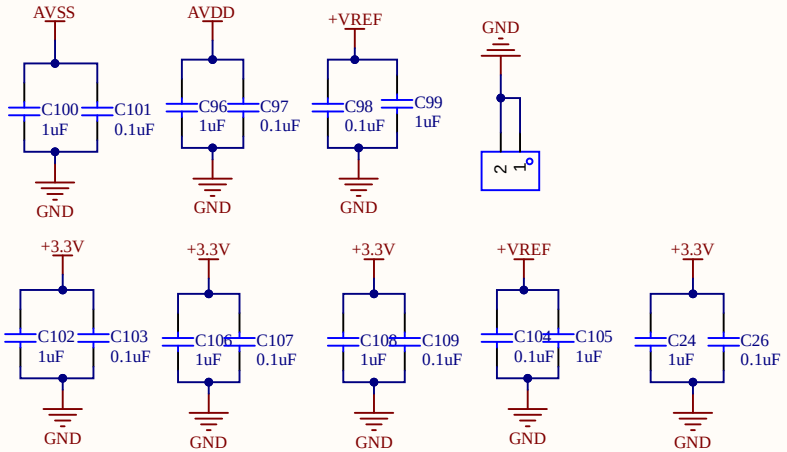


12位DAC—>1/1000　选择16位DAC

12位ACD—>1/1000　选择16位ADC

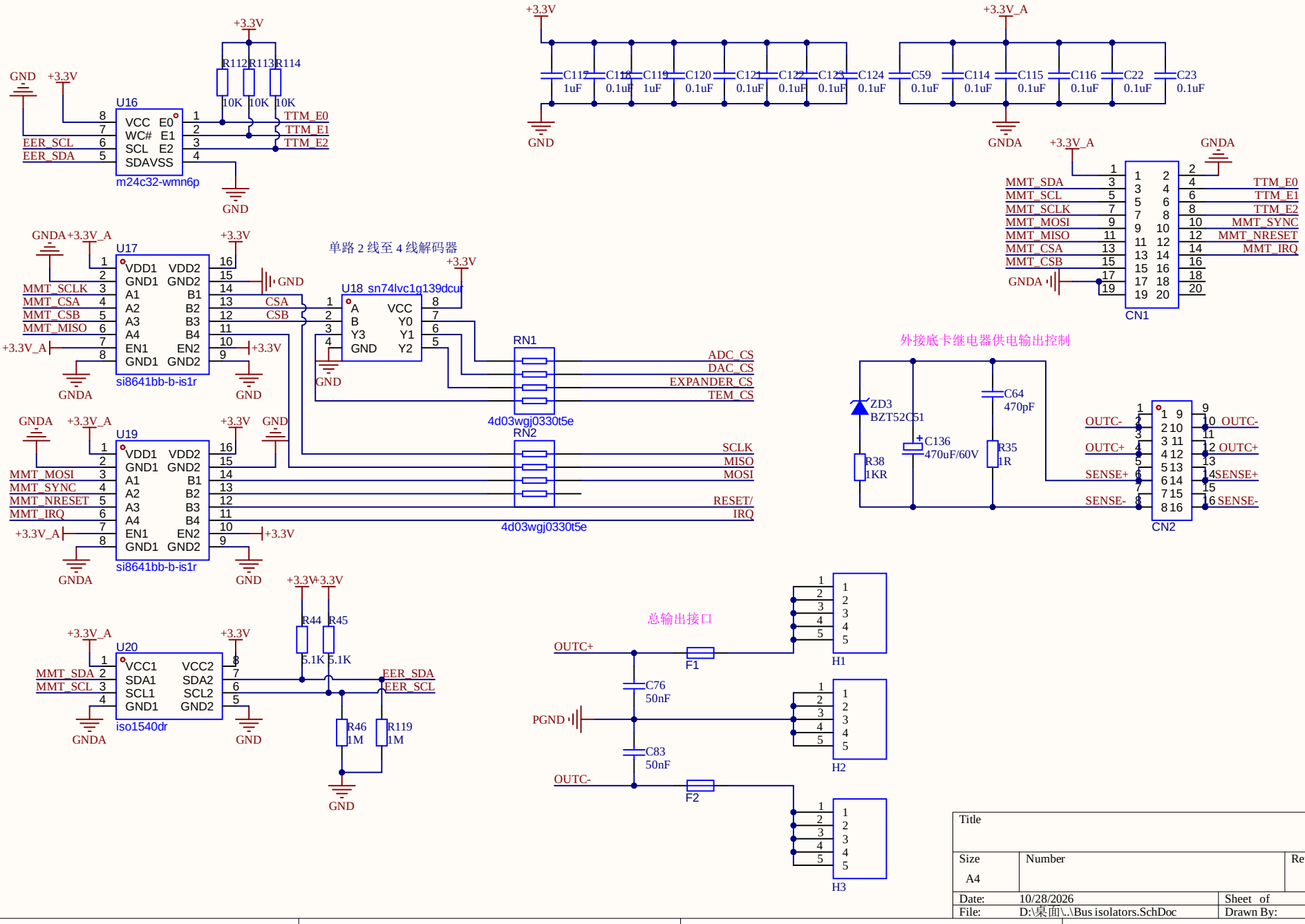
基准电压：+2.5V





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I2C 模块 ID EEPROM



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File: D:\桌面\..\Bus isolators.SchDoc		Drawn By: